

# Homework 2 – Sessions 3, 4 & 5

Functions · Function families · Proof techniques

## Part A · Session 3 – functions, composition and the unit circle

### 1 IMAGE AND PREIMAGE

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be given by  $f(x) = x^2 - 6x + 5$ .

- Find the image of  $f$  and prove your answer. State clearly which part of the argument shows the image is contained in your set, and which shows every element of your set is attained.
- Compute  $f^{-1}(\{0\})$  and  $f^{-1}(\{-5\})$ . Explain in one sentence what the second answer says about the relationship between the image and the codomain.

**SOLUTION** (a) Complete the square:  $f(x) = (x - 3)^2 - 4$ . Since  $(x - 3)^2 \geq 0$  for every real  $x$ , we get  $f(x) \geq -4$ , so the image is contained in  $[-4, \infty)$ . Conversely, given any  $y \geq -4$ , put  $x = 3 + \sqrt{y + 4}$  – which is real because  $y + 4 \geq 0$  – and then  $f(x) = y$ . Hence the image is exactly  $[-4, \infty)$ . The first half is the containment; the second is attainment.

(b)  $f(x) = 0$  means  $(x - 3)^2 = 4$ , so  $x - 3 = \pm 2$  and  $f^{-1}(\{0\}) = \{1, 5\}$ . For  $-5$ : since  $f(x) \geq -4 > -5$  for every  $x$ , no input maps to  $-5$ , so  $f^{-1}(\{-5\}) = \emptyset$ . An empty preimage is exactly what it means for a codomain element to lie outside the image – the codomain is declared, the image is computed.

## 2

## RESTRICTION AND INVERSE

Keep  $f(x) = x^2 - 6x + 5$ .

- Show that  $f : \mathbb{R} \rightarrow \mathbb{R}$  is not injective, with an explicit witness.
- Choose a domain  $D$  and codomain  $C$  making  $f : D \rightarrow C$  a bijection, and prove *both* halves.
- Write down the inverse function explicitly, and verify one composite.

**SOLUTION** (a)  $f(1) = 0 = f(5)$  while  $1 \neq 5$ , so  $f$  is not injective. (Any pair symmetric about  $x = 3$  works.)

(b) Take  $D = [3, \infty)$  and  $C = [-4, \infty)$ . *Injective*: suppose  $a, b \in D$  with  $f(a) = f(b)$ . Then  $(a - 3)^2 = (b - 3)^2$ , so  $|a - 3| = |b - 3|$ ; as  $a, b \geq 3$  both quantities are non-negative, giving  $a - 3 = b - 3$  and  $a = b$ . *Surjective*: given  $y \in C$ , the value  $x = 3 + \sqrt{y + 4}$  lies in  $D$  (the square root is  $\geq 0$ ) and satisfies  $f(x) = y$ , by part 1(a).

(c)  $f^{-1}(y) = 3 + \sqrt{y + 4}$  for  $y \geq -4$ . Check:  $f(f^{-1}(y)) = (3 + \sqrt{y + 4} - 3)^2 - 4 = (y + 4) - 4 = y$ .

The choice  $D = (-\infty, 3]$  with  $f^{-1}(y) = 3 - \sqrt{y + 4}$  is equally correct – but the domain must be stated, because “the function  $x^2 - 6x + 5$ ” names no single function.

## 3

## COMPOSITION

Let  $g(x) = 2x - 1$  and  $h(x) = x^2 + 1$ , both from  $\mathbb{R}$  to  $\mathbb{R}$ .

- Compute  $h \circ g$  and  $g \circ h$  as explicit formulas, and give a value of  $x$  at which they differ. What does this show about composition?
- Prove that  $g$  is a bijection and find  $g^{-1}$ .
- Prove or disprove:  $h \circ g$  is injective on  $\mathbb{R}$ .

**SOLUTION** (a)  $(h \circ g)(x) = (2x - 1)^2 + 1 = 4x^2 - 4x + 2$ , and  $(g \circ h)(x) = 2(x^2 + 1) - 1 = 2x^2 + 1$ . At  $x = 1$  the first gives 2 and the second gives 3. So composition is **not commutative**: the order of the maps matters.

(b) *Injective*:  $2a - 1 = 2b - 1 \Rightarrow a = b$ . *Surjective*: given  $y$ , take  $x = \frac{y+1}{2}$ . So  $g$  is a bijection with  $g^{-1}(y) = \frac{y+1}{2}$ .

(c) **Disproved**.  $(h \circ g)(x) = 4x^2 - 4x + 2 = (2x - 1)^2 + 1$ , so  $(h \circ g)(0) = 2 = (h \circ g)(1)$  while  $0 \neq 1$ . Note  $g$  is injective – so an injective first map does not make the composite injective. (The implication runs the other way: if  $h \circ g$  is injective, then  $g$  must be.)

#### 4 UNIT CIRCLE

Derive each value from the unit circle. For each, state the quadrant and the reference angle before giving the number; a recalled value with no working is not a derivation.

- $\cos \frac{5\pi}{6}$
- $\sin \frac{7\pi}{4}$
- $\tan \frac{2\pi}{3}$
- $\cos \frac{11\pi}{6}$

Then explain, in one sentence each, why **sin** is not injective on  $\mathbb{R}$  and what restriction of its domain makes it so.

**SOLUTION** (a) Quadrant II, reference  $\frac{\pi}{6}$ ; cosine is the  $x$ -coordinate and is negative there, so  $-\frac{\sqrt{3}}{2}$ .

(b) Quadrant IV, reference  $\frac{\pi}{4}$ ; sine is the  $y$ -coordinate and is negative there, so  $-\frac{\sqrt{2}}{2}$ .

(c) Quadrant II, reference  $\frac{\pi}{3}$ ;  $\tan = \frac{\sin}{\cos}$  is negative there, so  $-\sqrt{3}$ .

(d) Quadrant IV, reference  $\frac{\pi}{6}$ ; cosine is positive there, so  $\frac{\sqrt{3}}{2}$ .

**sin** is not injective because it is periodic:  $\sin(\theta + 2\pi) = \sin \theta$ , so for instance  $\sin 0 = \sin 2\pi = 0$  with  $0 \neq 2\pi$ . Restricting the domain to  $[-\frac{\pi}{2}, \frac{\pi}{2}]$  makes it injective (and onto  $[-1, 1]$ ), which is exactly how **arcsin** is defined.

#### 5 CLASSIFICATION

For each function, decide whether it is injective, surjective, both, or neither. Prove every positive claim; disprove every negative one with an explicit witness.

- $f: \mathbb{Z} \rightarrow \mathbb{Z}, f(n) = 3n - 7$
- $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^3 + x$
- $f: \mathbb{R} \setminus \{2\} \rightarrow \mathbb{R} \setminus \{1\}, f(x) = \frac{x+3}{x-2}$

**SOLUTION** (a) **Injective, not surjective.**  $3a - 7 = 3b - 7 \Rightarrow a = b$ . Not onto:  $f(n) = 0$  would need  $n = \frac{7}{3} \notin \mathbb{Z}$ ; more simply every output is  $\equiv -7 \equiv 2 \pmod{3}$ , so 0 is never attained.

(b) **Bijjective.** Injective: if  $a < b$  then  $a^3 < b^3$  and  $a < b$ , so  $f(a) < f(b)$  – a strictly increasing function is injective. Surjective:  $f$  is continuous with  $f(x) \rightarrow \pm\infty$  as  $x \rightarrow \pm\infty$ , so by the intermediate value theorem every real is attained.

(c) **Bijjective.** Injective:  $\frac{a+3}{a-2} = \frac{b+3}{b-2}$  cross-multiplies to  $(a+3)(b-2) = (b+3)(a-2)$ , i.e.  $ab - 2a + 3b - 6 = ab - 2b + 3a - 6$ , giving  $5b = 5a$  and  $a = b$ . Surjective: solving  $y = \frac{x+3}{x-2}$  gives  $x = \frac{2y+3}{y-1}$ , defined for every  $y \neq 1$ . This  $x$  cannot equal 2, since  $\frac{2y+3}{y-1} = 2$  would give  $2y + 3 = 2y - 2$ . Thus the preimage lies in the stated domain; this is exactly why 1 was removed from the codomain.

### 6 POLYNOMIAL STRUCTURE

Let  $p(x) = 2x^3 - 3x^2 - 11x + 6$ .

- State the degree and leading coefficient, and describe the end behaviour with a reason.
- Given that  $x = 3$  is a root, factor  $p$  completely over  $\mathbb{R}$  and list all roots.
- Sketch the graph, marking every  $x$ -intercept and the  $y$ -intercept.

**SOLUTION** (a) Degree **3**, leading coefficient **2**. The degree is odd and the leading coefficient positive, so  $p(x) \rightarrow -\infty$  as  $x \rightarrow -\infty$  and  $p(x) \rightarrow +\infty$  as  $x \rightarrow +\infty$ : down on the left, up on the right. For large  $|x|$  the term  $2x^3$  dominates the rest.

(b) Dividing by  $(x - 3)$  gives  $p(x) = (x - 3)(2x^2 + 3x - 2) = (x - 3)(2x - 1)(x + 2)$ . The roots are  $x = 3$ ,  $x = \frac{1}{2}$  and  $x = -2$ , each of multiplicity one.

(c)  $y$ -intercept  $p(0) = 6$ . The curve crosses at  $-2$ ,  $\frac{1}{2}$  and  $3$  – it crosses rather than touches at each, because every factor occurs once.

### 7 EXPONENTIAL EQUATION

Solve over the reals:

$$3^{2x} - 12 \cdot 3^x + 27 = 0.$$

Justify the substitution you use, and say why *both* resulting values must be checked before you convert back.

**SOLUTION** Put  $u = 3^x$ , so  $3^{2x} = (3^x)^2 = u^2$  and the equation becomes  $u^2 - 12u + 27 = 0$ . The substitution is legitimate because  $3^x$  is defined for every real  $x$  – but it takes **only positive values**, which is why each root must be tested: a non-positive root would correspond to no real  $x$  at all.

Factoring,  $u^2 - 12u + 27 = (u - 3)(u - 9)$ , so  $u = 3$  or  $u = 9$ . Both are positive, so both survive. Since  $3^t$  is one-to-one (base **3** is positive and  $\neq 1$ ),

$$3^x = 3 \Rightarrow x = 1, \quad 3^x = 9 = 3^2 \Rightarrow x = 2.$$

Check: at  $x = 1$ ,  $9 - 36 + 27 = 0$ ; at  $x = 2$ ,  $81 - 108 + 27 = 0$ . ✓

Solve exactly for  $x$ :

$$\log_3(x + 6) - \log_3(x - 2) = 2.$$

State the domain restriction *before* you combine the logarithms, and check your answer in the original equation.

**SOLUTION** *Domain first.* Both inputs must be positive:  $x + 6 > 0$  and  $x - 2 > 0$ , so  $x > 2$  (the stronger condition). Everything below happens on that domain.

By the quotient law,  $\log_3\left(\frac{x+6}{x-2}\right) = 2$ , so  $\frac{x+6}{x-2} = 3^2 = 9$ . Then  $x+6 = 9(x-2) = 9x-18$ , giving  $24 = 8x$  and  $x = 3$ .

$3 > 2$ , so the answer is admissible. Check:  $\log_3 9 - \log_3 1 = 2 - 0 = 2$ . ✓

Stating the domain is not bookkeeping: combining two logarithms into one *enlarges* the set of  $x$  the equation makes sense for, so an answer found after combining can easily be outside the original domain.

- Compute  $\log_8 32$  and  $\log_9 27$  exactly, showing the change-of-base step.
- Prove that  $\log_a b \cdot \log_b a = 1$  for all valid bases  $a, b$ , and state exactly which conditions on  $a$  and  $b$  the proof needs.

**SOLUTION** (a)  $\log_8 32 = \frac{\ln 32}{\ln 8} = \frac{5 \ln 2}{3 \ln 2} = \frac{5}{3}$ .  $\log_9 27 = \frac{\ln 27}{\ln 9} = \frac{3 \ln 3}{2 \ln 3} = \frac{3}{2}$ .

Both work because each pair is a power of a common base.

(b) By change of base,  $\log_a b = \frac{\ln b}{\ln a}$  and  $\log_b a = \frac{\ln a}{\ln b}$ , so the product is  $\frac{\ln b}{\ln a} \cdot \frac{\ln a}{\ln b} =$

1. The proof needs  $a, b > 0$  so the logarithms exist, and  $a \neq 1, b \neq 1$  so that  $\ln a \neq 0$  and  $\ln b \neq 0$  – otherwise a denominator vanishes.

Consider  $2^n$  and  $n^3$  for positive integers  $n$ .

- Find the smallest  $n_0$  such that  $2^n > n^3$  for every  $n \geq n_0$ . Give the two computations either side of the crossover, then give a short argument that the inequality continues to hold for every later  $n$ .
- Explain why checking a handful of values does not establish the claim for all  $n \geq n_0$ , and name the proof technique that would.

**SOLUTION** (a)  $n_0 = 10$ . At  $n = 9$ :  $2^9 = 512$  while  $9^3 = 729$ , so the inequality fails. At  $n = 10$ :  $2^{10} = 1024$  while  $10^3 = 1000$ , so it holds. Beyond  $n = 10$  the gap only widens, since doubling outpaces the cube's growth factor  $\left(\frac{n+1}{n}\right)^3$ , which drops below 2 once  $n \geq 4$ .

(b) A finite check covers finitely many  $n$ ; the claim is about infinitely many. Each verification is one instance of a universally quantified statement, and instances never exhaust a "for every" claim. The technique that does settle it is **induction** – prove the base case  $n = 10$ , then show  $2^k > k^3$  forces  $2^{k+1} > (k+1)^3$ . That is Session 6.

Bangkok's temperature over one day is modelled by

$$T(t) = 8 \sin\left(\frac{\pi(t-9)}{12}\right) + 22,$$

with  $T$  in  $^{\circ}\text{C}$  and  $t$  the hour,  $0 \leq t \leq 24$ .

- State the amplitude, midline and period, showing how you get the period from the coefficient.
- Give the maximum and minimum temperatures and the times at which they occur.
- The model has period 24 hours. What does that say about  $T(0)$  and  $T(24)$ , and is that physically reasonable?

**SOLUTION** (a) Amplitude  $|A| = 8$ , midline  $y = 22$ . The coefficient of  $t$  inside the sine is  $B = \frac{\pi}{12}$ , and the period is  $\frac{2\pi}{B} = \frac{2\pi}{\pi/12} = 24$  hours –  $B$  sits *inside* the function, so you divide  $2\pi$  by it rather than multiply.

(b) Sine reaches 1 when its input is  $\frac{\pi}{2}$ :  $\frac{\pi(t-9)}{12} = \frac{\pi}{2}$  gives  $t - 9 = 6$ , so  $t = 15$  and  $T = 8 + 22 = 30^{\circ}\text{C}$ . Sine reaches  $-1$  when the input is  $-\frac{\pi}{2}$ , giving  $t = 3$  and  $T = 22 - 8 = 14^{\circ}\text{C}$ . Maximum  $30^{\circ}\text{C}$  at 15:00, minimum  $14^{\circ}\text{C}$  at 03:00.

(c) Periodicity forces  $T(24) = T(0) = 8 \sin\left(-\frac{3\pi}{4}\right) + 22 \approx 16.3^{\circ}\text{C}$ . That is reasonable as a modelling assumption – midnight tonight should resemble midnight last night – but it is an assumption, not a fact about weather: a real cold front breaks it.

## 12 CONTRAPOSITIVE

Prove that for every integer  $n$ , if  $n^3$  is odd then  $n$  is odd.

- State the contrapositive of the claim.
- Prove the contrapositive, and say in one sentence why proving it settles the original.
- Would a direct proof have been harder? Say why.

**SOLUTION** (a) Contrapositive: for every integer  $n$ , if  $n$  is **even** then  $n^3$  is **even**.  
 (b) Let  $n$  be an arbitrary even integer. By definition  $n = 2k$  for some  $k \in \mathbb{Z}$ . Then  $n^3 = (2k)^3 = 8k^3 = 2(4k^3)$ , and  $4k^3 \in \mathbb{Z}$ , so  $n^3$  is even. ■ An implication and its contrapositive are true in exactly the same cases, so proving one proves the other.  
 (c) Yes. A direct proof starts from " $n^3$  is odd", i.e.  $n^3 = 2m + 1$ , and must extract information about  $n$  from a statement about its cube – there is no convenient factorisation to use. The contrapositive starts from  $n = 2k$ , which is a usable formula immediately. That is the general rule: contrapose when the negated conclusion is the more workable hypothesis.

## 13 CONTRADICTION

Prove that  $\sqrt{3}$  is irrational.

- Write out the proof by contradiction in full.
- Your argument will use the fact that if  $3 \mid a^2$  then  $3 \mid a$ . State where you used it, and why the analogous step would fail if  $3$  were replaced by  $4$ .

**SOLUTION** (a) Suppose  $\sqrt{3}$  is rational. Then  $\sqrt{3} = \frac{a}{b}$  with  $a \in \mathbb{Z}$ ,  $b \in \mathbb{N}$ , and  $a, b$  sharing no common factor greater than 1. Squaring,  $a^2 = 3b^2$ , so  $3 \mid a^2$  and therefore  $3 \mid a$ . Write  $a = 3a_1$ . Substituting,  $9a_1^2 = 3b^2$ , so  $b^2 = 3a_1^2$  and by the same fact  $3 \mid b$ . But then  $3$  divides both  $a$  and  $b$ , contradicting that they share no factor above 1. So  $\sqrt{3}$  is irrational. ■  
 (b) The fact is used twice: once to get  $3 \mid a$  from  $3 \mid a^2$ , and once to get  $3 \mid b$  from  $3 \mid b^2$ . It holds because  $3$  is **prime**. For  $4$  it fails:  $4 \mid 2^2$  but  $4 \nmid 2$ . And indeed the conclusion would be false –  $\sqrt{4} = 2$  is rational.

## 14 COUNTEREXAMPLE

Consider the claim: *for every non-negative integer  $n$ , the value  $n^2 - n + 41$  is prime.*

- Verify the claim for  $n = 0, 1, 2$  and  $3$ .
- The claim is nevertheless false. Find a counterexample and show it fails.
- In two or three sentences, say what this example shows about the relationship between evidence and proof.

**SOLUTION** (a)  $n = 0$ : 41.  $n = 1$ : 41.  $n = 2$ : 43.  $n = 3$ : 47. All four are prime.

(b) Take  $n = 41$ . Then

$$41^2 - 41 + 41 = 41^2 = 1681 = 41 \times 41,$$

which is composite. So the claim is false, and  $n = 41$  is a counterexample.

(c) The formula is prime for every  $n$  from 0 to 40 – forty-one consecutive successes – and still the universal claim is false. No finite amount of confirming evidence establishes a "for every" statement; it only fails to refute it. One counterexample, by contrast, settles the matter permanently. This is why the course insists on proof rather than checking.

## 15 QUANTIFIERS

Write the negation of each statement below in words, without using the phrase "it is not the case that".

- For every real  $x$ , there exists an integer  $n$  with  $n > x$ .
- There exists a real  $x$  such that for every real  $y$ ,  $xy = y$ .
- Decide whether each of the two original statements is true or false, and justify each answer in a sentence.

**SOLUTION** (a) There exists a real  $x$  such that for every integer  $n$ ,  $n \leq x$ .

(b) For every real  $x$ , there exists a real  $y$  with  $xy \neq y$ .

In both cases the rule is the same: each quantifier flips –  $\forall$  becomes  $\exists$  and  $\exists$  becomes  $\forall$ , in the order they appear – and the innermost property is negated.

(c) Both are **true**. For the first, given any real  $x$  the integer  $n = \lfloor x \rfloor + 1$  satisfies  $n > x$ , so no real number bounds all the integers from above. For the second,  $x = 1$  works, since  $1 \cdot y = y$  for every real  $y$ . Note that each statement is established in the shape its quantifiers demand: the  $\forall x \exists n$  statement needs an  $n$  built from an arbitrary  $x$ , while the  $\exists x \forall y$  statement needs only one explicit witness.